

MediTrust

1. Sprint 1 Objective

The main objective of Sprint 1 was to build a simple website prototype and successfully connect it to a database. We wanted to make sure that the frontend, backend, and PostgreSQL database were properly integrated and working together.

By the end of this sprint, the system should allow:

- User registration and login
- Basic risk calculation
- Storage of data in the database

2 What We Completed in This Sprint

During Sprint 1, we successfully implemented all planned features from the backlog.

First, we designed the frontend pages:

- Login page
- Signup page
- Risk assessment page

The UI allows users to create an account, log in, and enter patient details.

On the backend side, we implemented FastAPI and created API endpoints for:

- User registration
- User login
- Risk assessment

Passwords are securely hashed before storing in the database.

For the risk logic, we used a simple rule-based approach: (demo)

- If age is 50 or above → High risk (0.8)
- If age is below 50 → Low risk (0.3)

Even though this is a dummy logic, it demonstrates full functionality.

We also set up PostgreSQL using Docker and confirmed that:

- User accounts are stored in the database

- Assessment records are stored persistently
- Data remains available after restarting the server

Swagger documentation was also available to test APIs directly.

3. Demonstration of Functionality

The following workflow was tested successfully:

1. User creates a new account
2. Password is hashed and saved in PostgreSQL
3. User logs in with valid credentials
4. Patient details are entered
5. Risk level is calculated
6. Assessment result is saved in the database

We verified the data using SQL queries to confirm proper storage.

(Screenshots of login page, signup page, risk page, Database, Swagger docs, and GitHub board are attached.)

4. Project Management Tracking

We used a GitHub Kanban Board to manage tasks during Sprint 1.

Project Board Link:

<https://github.com/users/RavindraSSK/projects/7>

Tasks were moved through:

- Backlog
- Ready
- In Progress
- In Review
- Done

All Sprint 1 tasks were completed and moved to the Done column.

5. Changes from the Initial Plan

While working on Sprint 1, we made a few practical improvements:

- We added secure password hashing using bcrypt instead of storing plain text passwords.
- We used Docker for PostgreSQL instead of a local database installation.
- We added a database health endpoint for testing connectivity.

These improvements made the system more realistic and secure.

6. Challenges Faced

We encountered a few technical challenges:

- CORS errors when connecting frontend and backend
- Debugging 500 Internal Server errors
- bcrypt compatibility issue
- Docker configuration troubleshooting

However, after debugging and testing step by step, we resolved these issues successfully.

We now have:

- A working frontend prototype
- A FastAPI backend
- Secure authentication
- PostgreSQL database integration
- Full frontend-to-database communication

The project is now ready for further enhancements in Sprint 2.

Note: Respective screenshots are uploaded in screenshot folder